

Yihan Li

liyihan.xyz@gmail.com | +86 18615276438 | liyihan.xyz | github.com/HY-LiYihan

Education

Sun Yat-sen University	Sep 2023 – Jul 2027 (Expected)
<i>B.A. in English Language & Literature, School of Foreign Languages</i>	<i>GPA: 3.71/4.0</i>
<i>Minor in Computer Science & Technology, School of Computer Science and Engineering</i>	<i>GPA: 3.85/4.0</i>

Publications

Anticipating Contact for Visuo-Tactile Diffusion Policies	under review
<i>Third or subsequent author</i>	
Goldsmith: Gold-Loss-Guided Prompt Training for Agentic Annotation	under review
<i>First author</i>	
Unordered Landmark Visual Navigation	under review
<i>Third or subsequent author</i>	
OmniTrav: A Dataset and Benchmark for 360° Vision-based Traversability via Foundation Models	under review
<i>First author</i>	
RAPID Hand: Robust, Affordable, Perception-Integrated, Dexterous Manipulation Platform for Embodied Intelligence	NeurIPS 2025
<i>Third or subsequent author</i>	

Work Experience

HCL Lab (HKUST(GZ))	Jan 2026 – Present
<i>Research Assistant</i>	

- Researching dexterous-hand RL, Real-to-Sim pipelines, VLA models, and visuo-tactile policy learning.

Yun Drone	Jun 2025 – Mar 2026
<i>UAV Development Intern</i>	

- Handled UAV propulsion selection, formation-flight development, RL-based disturbance control, navigation/localization, and flight-test log analysis.

RAPID Lab (SYSU)	Nov 2024 – Jun 2025
<i>Research Assistant</i>	<i>Prof. Hui Cheng</i>

- Built dexterous-hand control, visuo-tactile alignment, MuJoCo Sim2Real validation, visual obstacle avoidance, and topological planning models.

RoboTech Association (SYSU)	Sep 2024 – Present
<i>President; Competition Team Captain</i>	

- Led 160+ members, daily training, competition teams, and robotics tutorials across mechanical, hardware, control, and algorithms.

Projects

RoboMaster Visual System Development (RMUL)	Mar 2025 – Jul 2025
<i>Vice Captain; Team Advisor; Algorithm Group Leader for Navigation and Vision</i>	

- Built Jetson/NUC vision with EKF auto-aiming (90%+ hit rate) and TensorRT/OpenVINO YOLO inference at 10 ms.
- Led navigation and vision algorithm planning, STM32 communication, and closed-loop deployment for competitive robotics.

RAPID Hand: Perception-Integrated Dexterous Manipulation Platform

Nov 2024 – Jun 2025

Research Assistant

- Implemented drivers, visuo-tactile fusion, MuJoCo Sim2Real training, and Diffusion Policy tasks with 50/50 success.
- Integrated 1 kHz hand control, wrist vision, fingertip tactile sensing, and joint proprioception under 7 ms synchronization latency.

OmniTrav: 360° Vision-based Traversability Benchmark

Jun 2025 – Mar 2026

First Author

- Proposed C2T RGB traversability and built a 4.7k-scene, 250k-frame, 577 GB automated foundation-model dataset.
- Evaluated pinhole, fisheye, and panoramic cameras; achieved panoramic obstacle detection F1 of 1.0 and pinhole median error of 0.27 m.

GenreShift and Rosetta: LLM Systems for Linguistic Research

Oct 2024 – Nov 2025

Project Leader

- Built GenreShift (corpus, LoRA, MoE, patent No. 2025105553019) and Rosetta (Streamlit/Docker annotation with Kimi/DeepSeek APIs).
- Supported interactive concept management, data import/export, and LLM-assisted academic discourse annotation.

Awards

- First Prize, National Level, China Robot and Artificial Intelligence Competition, 2025-08.
- Second Prize, National Level, National Undergraduate Electronics Design Contest, 2025-08.
- Second Prize, National Level, National Undergraduate Robotics Competition (RoboMaster / RoboCon), 2025.
- First Prize, National Level, National Intelligent Unmanned Systems Application Challenge, 2025-07.

Skills

Programming: ROS1/2, Python, C/C++, MATLAB, STM32/Arduino.

Robotics/ML: Visual navigation, 3D perception, dexterous manipulation, MuJoCo, TensorRT, OpenVINO, LLM annotation systems.

Engineering and Languages: Fusion 360, SolidWorks, EDA design; AOPA Multi-rotor VLOS Pilot; Mandarin (native), English (TEM-4), French.